

Seamless Integration of Practical Quantum Key Distribution in Future Passive Optical Networks via Adaptive Scheduling

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Abstract We investigate upstream DV-QKD in future passive optical networks under the finite-key regime, demonstrating that adaptive scheduling enables interference-reduced quantum-transmission windows and improves secret key rate, reach and scalability, especially in coexistence with XG-PON systems. ©2026 The Author(s)

Introduction

Quantum key distribution (QKD) represents the most promising solution for long-term secure communications and its integration into passive optical networks (PONs) is a key step toward end-to-end quantum security through the access segment of existing fibre infrastructures. In this context, QKD coexistence with legacy and emerging PON standards is challenged by splitter losses and spontaneous Raman scattering (SpRS) noise generated by high-power classical channels, which makes wavelength allocation and transmission conditions critical design elements. In our previous work, we estimated the performance of a discrete-variable (DV) QKD system in different PON integration scenarios in terms of quantum-available bandwidth and maximum reachable PON length [1]. More recently, adaptive scheduling has been proposed to enable an end-user to request QKD transmission cycles by temporarily silencing US classical traffic, thus improving the secret key rate (SKR) while preserving adequate quality of service for data transmission [2]. However, the previous scheduling-based analysis did not account for finite-key effects, even though, in practical QKD implementations, the time required to accumulate enough events for parameter estimation is crucial. Moreover, the study was limited to a single standard, while future access networks will progressively include heterogeneous coexistence with GPON, XG-PON and HS-PON standards [3]. In addition, the US adaptive scheduling not only reduces the SpRS noise affecting the QKD receiver, but also modifies the quantum available bandwidth (QAB).

In this paper, we analyse the performance of an upstream (US) DV-QKD system integrated into an adaptive PON scheduling scenario, including finite-key effects. Unlike previous work, we consider present and future deployments of emerging PON standards, analyse the full available spectrum under different operation conditions and explicitly evaluate the quantum-reserved cycles needed for

key distribution, quantifying their impact on the maximum reachable distance and the maximum supported number of users.

Network Architecture and Key Rate Estimation

We consider the integration scenario represented in Fig. 1. The optical line terminal (OLT) is connected to the optical splitter (S) through a feeder fibre of length L_f , while the splitter is connected to the optical network units (ONUs) through drop fibres of length L_d . We consider an US QKD channel to allow multiple end-users to share the expensive single-photon detectors located at the OLT. The PON infrastructure is particularly challenging for QKD integration as the transmittance of the quantum channel η is strongly affected by the loss introduced by the optical splitter and can be estimated as:

$$\eta = \frac{1}{N_{\text{user}}} e^{-\alpha_q(L_f+L_d)} \eta_d \quad (1)$$

where N_{user} is the number of users in the PON, defining the splitting ratio of the optical splitter, α_q is the attenuation coefficient of the quantum channel and η_d is the quantum efficiency of the single-photon detectors. Moreover, the QKD performance is degraded by the SpRS noise generated by the classical channels that in PON scenarios have standardised spectral allocations and launch powers. The SpRS photons overlap with the QKD channel increasing the background count rate as:

$$Y_0 = 2p_d + \frac{\lambda_q^3}{hc^2} N_R \Delta\nu \Delta t \eta_d \quad (2)$$

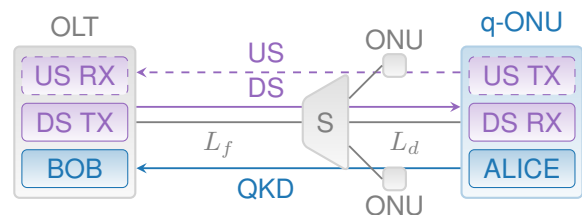


Fig. 1: Network architecture.

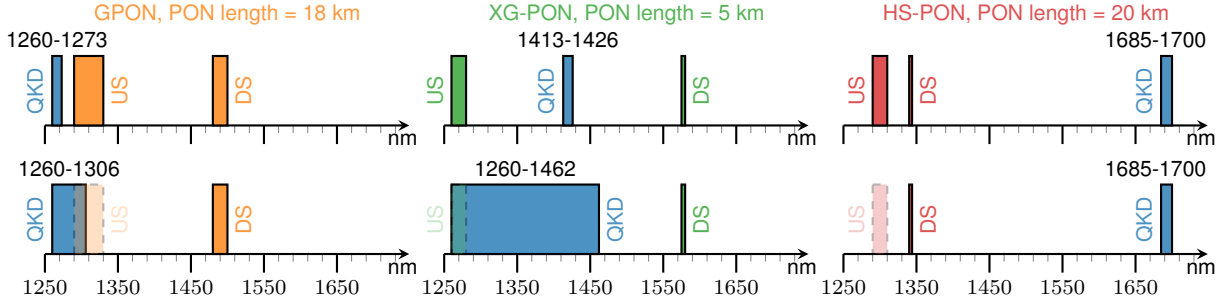


Fig. 2: Quantum available bandwidth in quantum-agnostic (top) and quantum-reserved (bottom) cycles at the maximum reachable distance in the quantum-agnostic scenario in a 16-user PON with $N = 10^{12}$.

where p_d is the dark count rate of the single-photon detectors, λ_q is the wavelength of the quantum channel, h is the Planck's constant, c is the speed of light, N_R is the total Raman noise spectral density, $\Delta\nu$ is the bandwidth of the bandpass filter at BOB side and Δt is the gate window. The total Raman noise is equal to the sum of the backward SpRS generated by the counter-propagating DS channels and the forward SpRS due to the co-propagation of the US channels and can be estimated according to the model and experimental parameters presented in [1] and validated with commercial devices in [4]. In PONs the US signal is characterized by burst transmission and uses a time division multiple access protocol. Therefore the standard dynamic bandwidth allocation (DBA) mechanism can be modified to avoid the SpRS generated by the US channel by reserving 125- μ s frame cycles to the quantum ONU (q-ONU) through an adaptive scheduling procedure [2]. In this scenario, the total Raman noise accounts only for the SpRS generated by the DS PON channel.

Based on the estimated η and Y_0 , we then calculate the SKR on the X basis in the finite-key regime according to [5]:

$$SKR = (s_{0,X} + s_{1,X}(1 - H(\phi_{1,X})) - \lambda_{EC} - 6 \log_2 \frac{21}{\epsilon_{sec}} - \log_2 \frac{2}{\epsilon_{cor}}) / N \quad (3)$$

where $s_{0,X}$ refers to the vacuum events, $s_{1,X}$ to the single-photon events, $\phi_{1,X}$ is the phase error rate, $H(x)$ is the binary Shannon entropy, λ_{EC} is the error correction leakage, N is the total number of transmitted states and the last two logarithmic terms account for the finite-key effects. A complete definition and derivation can be found in [5].

Results

We develop a design tool to analyse QKD performance in two scenarios: one with standard US DBA scheduling (quantum-agnostic cycles) and one with the adaptive scheduling (quantum-reserved cycles). The system parameters are re-

η_d	p_d	$\Delta\nu$	Δt	f_{EC}	ϵ_{sec}	ϵ_{cor}
0.25	10^{-6}	12.5 GHz	1 ns	1.05	10^{-10}	10^{-15}

Tab. 1: QKD system fixed parameters.

ported in Tab. 1, while the QKD intensities and probabilities are optimized for maximizing the resulting SKR in the 1260-1700 nm region with a 1-nm spectral resolution. The tool evaluates system performance as a function of the number of transmitted quantum states, the PON length and the number of users, in coexistence with different PON standards. Following [3], we analyse the integration with GPON, XG-PON and HS-PON whose spectral allocation and launch power are in accordance with the corresponding physical media dependent (PMD) layer specifications [6–8].

Firstly, we show that having interference-free QKD transmission windows frees spectral regions that would otherwise be occupied by the US channels. Fig. 2 represents the QABs with quantum-agnostic and quantum-reserved approaches at the maximum reachable distance achieved in the standard US DBA scheduling condition in a 16-user PON. In coexistence with GPON, the adaptive DBA scheduling expands the QAB by approximately 30 nm. Significant improvements are shown for the coexistence with XG-PON, which is the most challenging case in a quantum-agnostic scenario due to the presence of SpRS generated by both US and DS channels that force the QKD system to be placed in the E-band. With the adaptive DBA scheduling, the entire O-band becomes available for the QAB, as it is freed from the US channel and the associated SpRS Stokes components. On the other hand, the adaptive DBA

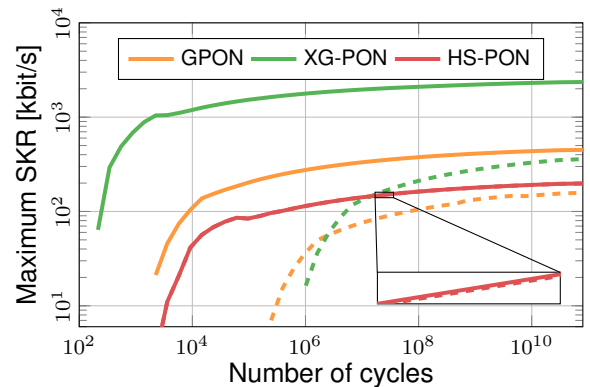


Fig. 3: Maximum achievable SKR as a function of the number of cycles in the quantum-agnostic (dashed) or quantum-reserved (solid) scenario in a 16-user PON. GPON @ 18 km, XGPON @ 5 km, HS-PON @ 20 km

scheduling has negligible impact on QKD performance in coexistence with HS-PON, since the allocation in opposite optical bands already provides sufficient noise isolation.

We then explicitly analyse the time dependence introduced by the finite-key regime. The time domain is divided into $125\text{-}\mu\text{s}$ long frame cycles, which correspond to blocks of 125000 transmitted quantum states for a 1-GHz QKD system. Fig. 3 shows the maximum achievable SKR for each PON standard in quantum-agnostic and quantum-reserved scenarios at the maximum distance reachable with standard DBA scheduling. As for the QAB, the greatest improvements are in coexistence with XG-PON, where the minimum number of cycles for extracting a key decreases from 10^6 to almost 10^2 ; the achievable SKR is also improved by at least one order of magnitude in a 5-km long PON. In case of GPON the adaptive scheduling allows to reduce the minimum number of cycles required for the key extraction by two order of magnitude. On the other hand, the use of the adaptive scheduling has a negligible impact when the system is integrated with the HS-PON standard as the maximum SKR overlaps in the two scenarios, as shown in the inset of Fig. 3.

While Fig. 3 shows the performance for a fixed number of users and PON lengths, we next investigate how the adaptive scheduling can extend the achievable network reach. In Fig. 4, we analyse a 64-user network by varying L_f from 1 to 19 km with a fixed $L_d = 1$ km for a maximum considered PON length equal to 20 km. With the standard DBA scheduling, the integration with HS-PON is the only scenario that saturates at 20 km after $4 \cdot 10^5$ cycles, whereas GPON and XG-PON stop at 18 and 4 km, respectively. With the adaptive scheduling, instead, a QKD system is able to distribute keys up to 20 km, coexisting with all three PON standards requesting to reserve between $1 \cdot 10^5$ to $4 \cdot 10^6$ cycles.

Finally, the maximum supported number of users also depends on the considered scenario

and on the available transmission time. Increasing the number of ONUs connected to the same OLT increases the splitting ratio, thereby reducing the transmittance of the quantum channel and modifying the SpRS generation. Considering a 20-km long PON, the results presented in Fig. 5 show similar behaviors to the ones shown in Fig. 4. The greatest improvements are obtained for the XG-PON standard, when the presence of interference-free windows enables key generation up to the maximum considered number of users (i.e., 128), whereas 20 km are not reachable even in a point-to-point (1-user PON) scenario with the standard DBA scheduling. In coexistence with GPON, the network architecture is limited to 42 user without adaptive scheduling while it can support 113 ONUs with quantum-reserved cycles. As for the previous figures, the integration with HS-PON is not affected by the adoption of the adaptive scheduling.

Conclusions

We presented a finite-key analysis of US DV-QKD coexistence with GPON, XG-PON and HS-PON under both quantum-agnostic and quantum-reserved cycles allocation strategies, quantifying the impact on achievable SKR, network reach and supported splitting ratio. The results highlight that adaptive scheduling is crucial for coexistence with XG-PON, whereas negligible advantages are obtained for HS-PON.

Despite the high SKR values, finite-key operation still requires the accumulation of a sufficiently large number of transmitted states for accurate parameter estimation. QKD transmission should therefore be scheduled dynamically and opportunistically, also leveraging ONU sleep-mode periods. This further supports the adoption of an asynchronous mechanism between key generation and key consumption, since the large number of reserved cycles needed to saturate performance makes fully synchronous and request-triggered operation inefficient for both quantum and classical systems.

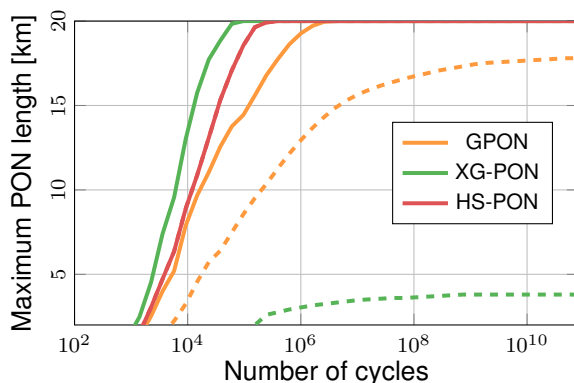


Fig. 4: Maximum achievable length as a function of the number of cycles in the quantum-agnostic (dashed) or quantum-reserved (solid) scenario in a 64-user PON.

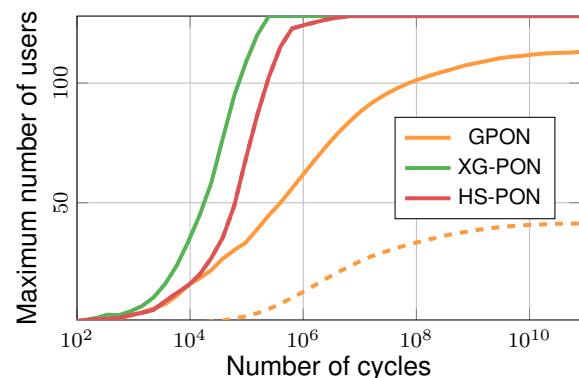


Fig. 5: Maximum number of users as a function of the number of cycles in the quantum-agnostic (dashed) or quantum-reserved (solid) scenario at 20 km.

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